

EC303 · SVNIT SURAT · THE TU CHAHIYE QUIZ, ON PAPER

DIGITAL SIGNAL PROCESSING

40 QUESTIONS

90 MARKS

45 MINUTES

ANSWER KEY AT THE BACK

How the marks work. Ten easy questions at 1 mark, ten medium at 2, and twenty hard at 3 — ninety in total, two thirds of them in the hard twenty. One option is right in every question. **The answer key, with a short explanation for every question, starts on the last section of this document.**

40

QUESTIONS

90

MARKS

45:00

ON THE CLOCK

1

ATTEMPT ONLINE

- 01** What does the unit impulse $\delta(n)$ equal? 1 MARK
- A 1 for every n B 1 at $n = 0$ and 0 everywhere else
C 0 at $n = 0$ and 1 everywhere else D n for every n
- 02** An LTI system is completely described by which single function? 1 MARK
- A Its step response B Its impulse response $h(n)$
C Its input signal D Its sampling rate
- 03** If $x(n)$ has length 4 and $h(n)$ has length 3, how long is their linear convolution? 1 MARK
- A 3 B 4
C 6 D 7
- 04** What does ROC stand for in z-transform work? 1 MARK
- A Rate of Change B Region of Convergence
C Range of Coefficients D Root of Characteristic
- 05** How many complex multiplications does a direct N-point DFT need? 1 MARK
- A N B $N \log_2 N$
C N squared D $N/2$
- 06** What is the twiddle factor $W(N)$ in the DFT? 1 MARK
- A e raised to $(-j2\pi/N)$ B e raised to $(+j2\pi N)$
C $2\pi/N$ D $N/2\pi$

- 07** Aliasing happens when... 1 MARK
- A the sampling rate is more than twice the highest frequency B the sampling rate is less than twice the highest frequency
- C the signal is quantized too finely D the signal is periodic
- 08** For a discrete-time LTI system, what is the condition for BIBO stability? 1 MARK
- A $h(n)$ must be finite in length B The sum of $|h(n)|$ over all n must be finite
- C $h(n)$ must be causal D $h(n)$ must be real
- 09** In a radix-2 FFT of length N , how many stages are there? 1 MARK
- A N B $N/2$
- C $\log_2 N$ D $2N$
- 10** What does a causal system mean? 1 MARK
- A Its output depends only on present and past inputs B Its output depends only on future inputs
- C It has no memory D Its impulse response is symmetric
- 11** Is $x(n) = \cos(\pi n/8 + 0.2)$ periodic, and if so what is its fundamental period? 2 MARKS
- A Not periodic B Periodic, $N = 8$
- C Periodic, $N = 16$ D Periodic, $N = 4$
- 12** Is $x(n) = e^{jn}$ periodic? 2 MARKS
- A Yes, $N = 1$ B Yes, $N = 2\pi$
- C Yes, $N = 6$ D No
- 13** Convolve $x(n) = \{1, 2, 3\}$ with $h(n) = \{1, 1\}$. 2 MARKS
- $x(n) = 1, 2, 3 \quad (n = 0, 1, 2)$
 $h(n) = 1, 1 \quad (n = 0, 1)$
- A $\{1, 3, 5, 3\}$ B $\{1, 2, 3, 1\}$
- C $\{1, 3, 5\}$ D $\{1, 2, 5, 3\}$
- 14** What is the z-transform of $x(n) = a^n \cdot u(n)$, and what is its ROC? 2 MARKS
- A $1/(1 - a \cdot z^{-1})$, ROC $|z| > |a|$ B $1/(1 - a \cdot z^{-1})$, ROC $|z| < |a|$
- C $1/(1 + a \cdot z^{-1})$, ROC $|z| > |a|$ D $a/(z - 1)$, ROC all z
- 15** A causal system has poles at $z = 0.5$ and $z = 2$. Is it stable? 2 MARKS
- A Yes, because one pole is inside the unit circle B No, because the pole at $z = 2$ is outside the unit circle
- C Yes, because it is causal D Cannot be decided without the zeros

- 16** In a decimation-in-time radix-2 FFT, what order are the input and the output in? 2 MARKS
- A Input natural, output bit-reversed B Input bit-reversed, output natural
C Both bit-reversed D Both natural
- 17** For an 8-point FFT, what is the bit-reversed input order? 2 MARKS
- A 0, 1, 2, 3, 4, 5, 6, 7 B 0, 4, 2, 6, 1, 5, 3, 7
C 7, 6, 5, 4, 3, 2, 1, 0 D 0, 2, 4, 6, 1, 3, 5, 7
- 18** What is the value of the twiddle factor $W(8)^4$? 2 MARKS
- A +1 B -1
C +j D -j
- 19** When is circular convolution of two sequences equal to their linear convolution? 2 MARKS
- A Always B Never
C When the DFT length N is at least $L_1 + L_2 - 1$ D When both sequences are the same length
- 20** What does the DFT of the shifted impulse $x(n) = \delta(n - n_0)$ look like? 2 MARKS
- A A single impulse at $k = n_0$ B A constant of magnitude 1 with a linear phase $-2\pi k \cdot n_0 / N$
C Zero everywhere D A constant N with zero phase
- 21** A 4-point signal has $x(0) = -2$, $x(1) = 1$, $x(2) = 1$, $x(3) = 2$. What is $X(1)$? 3 MARKS
- $x(n) = -2, 1, 1, 2$ for $n = 0..3$
 $X(k) = \sum x(n) (-j)^{nk}$
- A $-3 + j$ B $-3 - j$
C -4 D 2
- 22** For that same 4-point signal, the question asks for $X(0)$ through $X(7)$. What are $X(4)$ to $X(7)$? 3 MARKS
- A All zero B The same as $X(0)$ to $X(3)$, because the DFT is periodic with period N
C The negatives of $X(0)$ to $X(3)$ D Undefined — a 4-point DFT has only four values
- 23** An LTI system is described by $y(n) - \frac{1}{2} \cdot y(n-1) = x(n)$. What is the magnitude of its frequency response at $\omega = 0$? 3 MARKS
- A 0.5 B 1.0
C 1.5 D 2.0

- 24** For the same system, what is $|H(e^{j\omega})|$ at $\omega = \pi$? 3 MARKS
- A $2/3$ B $1/2$
 C 2 D 1
- 25** How many complex multiplications does a 1024-point radix-2 FFT need, and how does that compare with the direct DFT? 3 MARKS
- A 5120, about 205 times fewer B 10240, about 102 times fewer
 C 1024, about 1024 times fewer D 512, about 2048 times fewer
- 26** A causal LTI system has $H(z) = 1/(1 - 0.5z^{-1})(1 - 2z^{-1})$. What are its ROC, its stability, and can it be made stable? 3 MARKS
- A ROC $|z| > 2$, unstable; a stable version exists but is non-causal B ROC $|z| > 0.5$, stable
 C ROC $0.5 < |z| < 2$, stable and causal D ROC $|z| < 0.5$, unstable
- 27** An accumulator is defined by $y(n) = x(n) + y(n-1)$. Is it stable? 3 MARKS
- A Yes, because its impulse response is bounded B No, because $h(n) = u(n)$ is not absolutely summable
 C Yes, because it is causal D Only if the input is bounded
- 28** A pressure gauge modelled as an LTI system gives $(1 - e^{-t} - t \cdot e^{-t})u(t)$ for a unit step input. What is its impulse response? 3 MARKS
- A $(1 - e^{-t})u(t)$ B $t \cdot e^{-t} \cdot u(t)$
 C $e^{-t} \cdot u(t)$ D $(1 + t)e^{-t} \cdot u(t)$
- 29** Explain why decimation-in-frequency splits differently from decimation-in-time. 3 MARKS
- A DIT splits the input into even and odd samples; DIF splits it into first and second halves B DIT splits the output; DIF splits the input
 C They are the same algorithm with different names D DIT works only for real inputs
- 30** How is the inverse DFT computed using a forward FFT routine? 3 MARKS
- A Run the FFT backwards through the butterflies B Conjugate the input, run the forward FFT, conjugate the result and divide by N
 C Replace $W(N)$ with $W(N)^{*2}$ and run the FFT D Reverse the input sequence and run the FFT

- 31** Why does the FFT store $2N$ locations for data plus $N/2$ for twiddle factors, and what is in-place computation? 3 MARKS
- A Because it needs two copies of the data; in-place means using extra memory B Because the N complex values need real and imaginary parts; in-place means each butterfly writes its results back over the two inputs it just read
- C Because the twiddles double each stage; in-place means storing every stage D Because it computes forwards and backwards at once
- 32** For a radix-3 FFT with $N = 9$, how many stages are needed, and what is the general butterfly size? 3 MARKS
- A 2 stages, 3-point butterflies B 3 stages, 2-point butterflies
- C 9 stages, 3-point butterflies D 2 stages, 9-point butterflies
- 33** Find the particular solution of $y(n) - \frac{1}{2}y(n-1) = x(n)$ when $x(n) = 2^{n/2} \cdot u(n)$. 3 MARKS
- A $y(n) = (4/3) \cdot 2^{n/2}$ B $y(n) = 2^{n/2}$
- C $y(n) = (2/3) \cdot 2^{n/2}$ D $y(n) = (1/2) \cdot 2^{n/2}$
- 34** Which of these describes the difference between an energy signal and a power signal? 3 MARKS
- A Energy signals are finite in length, power signals are infinite B Energy signals have finite total energy and zero average power; power signals have infinite energy and finite non-zero average power
- C Energy signals are periodic, power signals are not D Energy signals are discrete, power signals are continuous
- 35** What is the ROC of the z-transform of a finite-length sequence $x(n) = \{1, 2, 3\}$ for $n = 0, 1, 2$? 3 MARKS
- A $|z| > 3$ B $|z| < 1$
- C The entire z-plane except $z = 0$ D The entire z-plane except $z = \infty$
- 36** A discrete-time system has the input–output relation $y(n) = x(-n)$. Is it time-invariant? 3 MARKS
- A Yes, because it is linear B No — delaying the input and reflecting do not commute
- C Yes, because reflection preserves energy D Only if $x(n)$ is even
- 37** Why is a single sampling frequency of 1 kHz unable to distinguish 300 Hz from 700 Hz? 3 MARKS
- A Because 700 Hz is above the 500 Hz folding frequency, so it aliases down to $1000 - 700 = 300$ Hz B Because both are below 1 kHz
- C Because the quantizer cannot resolve them D Because 700 Hz is not a multiple of 1 kHz
- 38** Two systems with impulse responses $h_1(n)$ and $h_2(n)$ are connected in cascade, then that whole thing is put in parallel with $h_3(n)$. What is the overall impulse response? 3 MARKS
- A $h_1(n) \cdot h_2(n) + h_3(n)$ B $h_1(n) * h_2(n) + h_3(n)$
- C $h_1(n) + h_2(n) + h_3(n)$ D $(h_1(n) + h_2(n)) * h_3(n)$

39 Zero-padding an 8-point sequence to 64 points before taking the DFT does what? 3 MARKS

- A Improves the true frequency resolution eightfold
- B Gives a finer-looking spectrum by interpolating between the same 8 genuine points, without adding new information
- C Removes spectral leakage completely
- D Changes the underlying signal

40 What does the convolution property of the z-transform say, and why is it useful? 3 MARKS

- A Convolution in time becomes addition in z
- B Convolution in time becomes multiplication in z, turning a sum-of-products into an algebra problem
- C Multiplication in time becomes convolution in z, which is harder
- D It only holds for causal sequences

The answer key

The letter, then why. Read the explanation even where you got it right — that is where most of the revision value is.

- 01 **B** $\delta(n) = 1$ when $n = 0$ and 0 otherwise. It matters because any sequence can be written as a sum of shifted, scaled impulses — and that fact is exactly what makes convolution work for LTI systems.
- 02 **B** Once you know $h(n)$ you know everything: the output for any input is $y(n) = x(n) * h(n)$. Linearity and time invariance are what let one response stand in for all of them.
- 03 **C** $\text{Length} = L_1 + L_2 - 1 = 4 + 3 - 1 = 6$. Always compute this before you start — it tells you how many numbers you are looking for, and half the marks in a convolution question come from getting the length right.
- 04 **B** Region of Convergence — the set of z values for which the sum actually converges. It is not optional extra information: the same algebraic $X(z)$ with two different ROCs describes two completely different sequences, one causal and one anti-causal.
- 05 **C** N^2 complex multiplications and $N(N-1)$ complex additions, because each of the N outputs is a sum over all N inputs. The radix-2 FFT cuts this to $(N/2) \cdot \log_2(N)$, and that single change is what made real-time spectral analysis possible.
- 06 **A** $W(N) = e^{-j2\pi/N}$, the complex N th root of unity. The DFT is $X(k) = \sum x(n) \cdot W(N)^{nk}$, and the FFT exists because the powers of $W(N)$ repeat — the symmetry $W(N)^{k+N/2} = -W(N)^k$ is what lets half the work be reused.
- 07 **B** Below the Nyquist rate, a high-frequency component folds back and appears as a lower frequency that was never in the original. Once it has happened it is permanent — no filter afterwards can separate the impostor from the genuine low frequency.
- 08 **B** Absolute summability: $\sum |h(n)| < \infty$. In the z -domain the same condition says the ROC must include the unit circle. For a causal system that means every pole lies strictly inside the unit circle.
- 09 **C** $\log_2(N)$ stages, each with $N/2$ butterflies. For $N = 8$ that is 3 stages of 4 butterflies; for $N = 1024$ it is 10 stages of 512. The total $(N/2) \cdot \log_2(N)$ multiplications is where the whole saving comes from.
- 10 **A** The output at time n may use $x(n)$, $x(n-1)$, $x(n-2)$... but never $x(n+1)$. Equivalently $h(n) = 0$ for $n < 0$. Any system that has to run in real time must be causal, because the future has not arrived yet.
- 11 **C** A discrete sinusoid $\cos(\omega_0 \cdot n + \phi)$ is periodic only if $\omega_0/2\pi$ is rational. Here $\omega_0 = \pi/8$, so $N = 2\pi k/\omega_0 = 2\pi k/(\pi/8) = 16k$, and the smallest whole-number answer is $N = 16$. The phase 0.2 shifts the wave but cannot change its period.
- 12 **D** Here $\omega_0 = 1$, so periodicity would need $N = 2\pi k/1 = 2\pi k$ to be a whole number for some integer k — impossible, because π is irrational. The continuous-time signal $e^{j\omega_0 t}$ is perfectly periodic; sampling it destroys the periodicity. This is one of the real differences between the discrete and continuous worlds.
- 13 **A** Length $3 + 2 - 1 = 4$. Each output is a sum of products whose indices add up to n : $y(0) = 1 \cdot 1 = 1$; $y(1) = 1 \cdot 1 + 2 \cdot 1 = 3$; $y(2) = 2 \cdot 1 + 3 \cdot 1 = 5$; $y(3) = 3 \cdot 1 = 3$. Free check: $\sum y = 12$ must equal $(\sum x)(\sum h) = 6 \times 2 = 12$.
- 14 **A** $X(z) = \sum a \cdot z^{-n} = \sum (a \cdot z^{-1})^n$, a geometric series that converges when $|a \cdot z^{-1}| < 1$, that is $|z| > |a|$ — the outside of a circle through the pole. The same $X(z)$ with ROC $|z| < |a|$ belongs to the anti-causal sequence $-\sum a \cdot z^{-n} \cdot u(-n-1)$, which is why the ROC must always be stated.
- 15 **B** For a causal system the ROC is everything outside the outermost pole, so here it is $|z| > 2$ — which does not contain the unit circle, so the system is unstable. A causal system is stable only when every pole lies strictly inside the unit circle. Zeros never affect stability.
- 16 **B** DIT splits the input into even and odd samples, so the input has to be shuffled into bit-reversed order and the output comes out naturally ordered. DIF is the mirror image: natural in, bit-reversed out. Both need exactly the same number of operations.

- 17 **B** Write each index in 3 bits and reverse the bits: $0 = 000 \rightarrow 000 = 0$; $1 = 001 \rightarrow 100 = 4$; $2 = 010 \rightarrow 010 = 2$; $3 = 011 \rightarrow 110 = 6$; and so on. The sequence is 0, 4, 2, 6, 1, 5, 3, 7. It is not the same as taking evens then odds once — that is only the first split.
- 18 **B** $W(8)^{4/8} = e^{-j2\pi \cdot 4/8} = e^{-j\pi} = -1$. In general $W(N)^{N/2} = -1$, and that single identity is the symmetry the whole butterfly is built on: it turns a multiplication into a sign change.
- 19 **C** Circular convolution wraps anything past N back onto the start, which corrupts the first few samples — time-domain aliasing. Zero-padding both sequences to $N \geq L_1 + L_2 - 1$ leaves room for the whole result, so nothing wraps and the two agree exactly.
- 20 **B** $X(k) = e^{-j2\pi k \cdot n_0/N}$. The magnitude is 1 at every k — a single sample is spread over all frequencies equally — while the shift appears entirely in the phase, as a straight line. This is the time-shift property: shifting in time multiplies by a complex exponential in frequency.
- 21 **A** With $N = 4$ the twiddle $W(4)^{nk} = (-j)^{nk}$. $X(1) = -2 \cdot 1 + 1 \cdot (-j) + 1 \cdot (-j)^2 + 2 \cdot (-j)^3 = -2 - j + 1(-1) + 2(j) = -3 + j$. For the record the rest are $X(0) = 2$, $X(2) = -4$, $X(3) = -3 - j$, and $X(0)$ and $X(3)$ are conjugate-symmetric partners because $x(n)$ is real.
- 22 **B** The DFT is periodic in k with period N , so $X(k + N) = X(k)$. With $N = 4$ that gives $X(4) = X(0) = 2$, $X(5) = X(1) = -3 + j$, $X(6) = X(2) = -4$ and $X(7) = X(3) = -3 - j$. The trap in the question is that being asked for eight values does not make it an 8-point DFT.
- 23 **D** Take the z -transform: $H(z) = 1/(1 - 0.5z^{-1})$, so $H(e^{j\omega}) = 1/(1 - 0.5e^{-j\omega})$. At $\omega = 0$, $e^{-j\omega} = 1$, giving $H = 1/(1 - 0.5) = 2$. In general $|H(e^{j\omega})| = 1/\sqrt{1.25 - \cos \omega}$, which is largest at $\omega = 0$ and smallest at $\omega = \pi$ — so this is a low-pass system with a DC gain of 2.
- 24 **A** At $\omega = \pi$, $e^{-j\pi} = -1$, so $H = 1/(1 + 0.5) = 2/3 \approx 0.667$. Check against the general form: $1/\sqrt{1.25 - \cos \pi} = 1/\sqrt{2.25} = 1/1.5 = 0.667$. Gain 2 at DC falling to 0.667 at the highest discrete frequency confirms the low-pass character.
- 25 **A** FFT: $(N/2) \cdot \log_2(N) = 512 \times 10 = 5120$. Direct DFT: $N^2 = 1,048,576$. The ratio is $1,048,576 / 5120 = 204.8$, so roughly a 205-fold saving. The saving grows with N : at $N = 8$ it is only about 5, at $N = 512$ it is 114.
- 26 **A** Poles are at 0.5 and 2. Causal means the ROC is outside the outermost pole, $|z| > 2$, which excludes the unit circle — unstable. The same $H(z)$ with ROC $0.5 < |z| < 2$ does include the unit circle, so it is stable, but that ROC is an annulus and therefore two-sided, i.e. non-causal. You can have causal or stable here, not both.
- 27 **B** The impulse response is the unit step $u(n)$, and $\sum |u(n)| = 1 + 1 + 1 + \dots$ diverges, so BIBO stability fails. In the z -domain $H(z) = 1/(1 - z^{-1})$ has its pole exactly on the unit circle at $z = 1$ — not inside it. A bounded input such as $u(n)$ gives the unbounded output $(n+1)u(n)$, which is the definition being violated.
- 28 **B** The impulse response is the derivative of the step response. Differentiating: $d/dt[1 - e^{-t} - t \cdot e^{-t}] = e^{-t} - (e^{-t} - t \cdot e^{-t}) = t \cdot e^{-t}$. So $h(t) = t \cdot e^{-t} \cdot u(t)$. The same idea in the transform domain: $H(s) = s \cdot (\text{step response})$, and the step response $1/s - 1/(s+1) - 1/(s+1)^2$ times s gives $1/(s+1)^2$, which inverts to $t \cdot e^{-t}$.
- 29 **A** DIT divides the input sequence into its even-indexed and odd-indexed samples, which is why the input must be bit-reversed. DIF divides the input into its first $N/2$ and last $N/2$ samples and splits the OUTPUT into even and odd frequency bins, so the output comes out bit-reversed. The butterfly is also arranged differently: DIT multiplies by the twiddle before the add-subtract, DIF after.
- 30 **B** The IDFT differs from the DFT only by the sign in the exponent and a $1/N$ factor. Since conjugating flips that sign, $x(n) = (1/N) \cdot \text{conj}(\text{FFT}(\text{conj}(X(k))))$. One routine then serves both directions, which is why a hardware FFT block never needs a separate inverse unit.
- 31 **B** N complex data values need $2N$ real storage locations, and the $N/2$ twiddle factors are counted alongside them, giving the textbook figure of $5N/2$. In-place means a butterfly reads two values, computes its two outputs, and writes them straight back into the same two locations — so no extra array is needed for the intermediate stages. Bit reversal is the price paid for that.

- 32 **A** Radix-3 needs $N = 3^v$, and the number of stages is $v = \log_3(N) = \log_3(9) = 2$. Each butterfly now handles three inputs rather than two, and each stage contains $N/3 = 3$ of them. The same divide-and-conquer idea drives every radix — split the DFT into smaller DFTs and reuse the twiddle symmetry.
- 33 **A** For a forcing function 2^n , try a particular solution $y(n) = K \cdot 2^n$. Substituting: $K \cdot 2^n - \frac{1}{2} \cdot K \cdot 2^{n-1} = 2^n$, so $K \cdot 2^n - (K/4) \cdot 2^n = 2^n$, giving $K(1 - \frac{1}{4}) = 1$ and $K = 4/3$. So $y(n) = (4/3) \cdot 2^n$ for $n \geq 0$. The complete solution would add the homogeneous term $C \cdot (\frac{1}{2})^n$, fixed by the initial condition.
- 34 **B** A decaying signal such as $(\frac{1}{2})^n \cdot u(n)$ has a finite energy sum and therefore zero average power — an energy signal. A signal that never dies away, such as a sinusoid or a unit step, has infinite energy but a finite non-zero mean square — a power signal. Every periodic signal is a power signal, but a signal can be a power signal without being periodic. No signal can be both.
- 35 **C** $X(z) = 1 + 2z^{-1} + 3z^{-2}$, a polynomial in z^{-1} with no poles anywhere finite except at $z = 0$, where z^{-2} blows up. So the ROC is everywhere except the origin. If the sequence also had samples at negative n it would contain positive powers of z , and $z = \infty$ would have to be excluded too.
- 36 **B** Delay first, then reflect: $x(n - n_0)$ becomes $x(-(n - n_0))$. Reflect first, then delay: $x(-n)$ becomes $x(-(n - n_0)) = x(-n + n_0)$. Those are different, so the system is time-varying. It is linear, though — folding is a linear operation. Linearity and time invariance are independent properties and must be tested separately.
- 37 **A** The folding frequency is $f_s/2 = 500$ Hz. Anything above it folds back to $|f_s - f| = 1000 - 700 = 300$ Hz, so after sampling the two are literally the same sequence of numbers. The only cure is an anti-aliasing low-pass filter before the sampler, because once the samples exist the information is gone.
- 38 **B** Cascade means convolution of the impulse responses, parallel means addition, so $h(n) = h_1(n) * h_2(n) + h_3(n)$. In the z -domain the same statement is $H(z) = H_1(z) \cdot H_2(z) + H_3(z)$ — which is usually far easier to work with, and is the main reason for transforming in the first place.
- 39 **B** True resolution is set by how long you actually observed the signal, not by how many zeros you bolt on afterwards. Padding interpolates the same underlying DTFT more densely, so the plot looks smoother and peaks are easier to locate — but two tones that were unresolvable in 8 genuine samples stay unresolvable. To get real resolution you need more real data.
- 40 **B** $y(n) = x(n) * h(n)$ becomes $Y(z) = X(z) \cdot H(z)$, with an ROC that contains at least the intersection of the two. It is useful because convolving two long sequences by hand is a bookkeeping exercise, while multiplying two rational functions and inverting by partial fractions is ordinary algebra — and it is also how cascades of systems are analysed.